

Intermediate Microeconomics

Week 1(b): Math Review

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Use of mathematics in economics

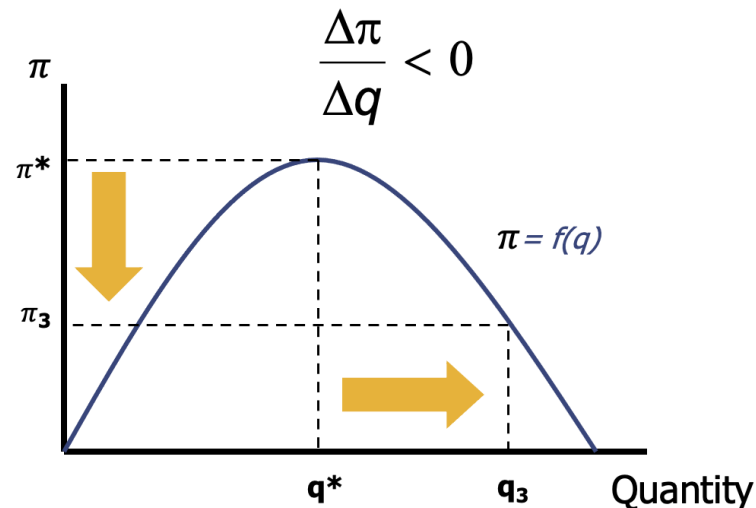
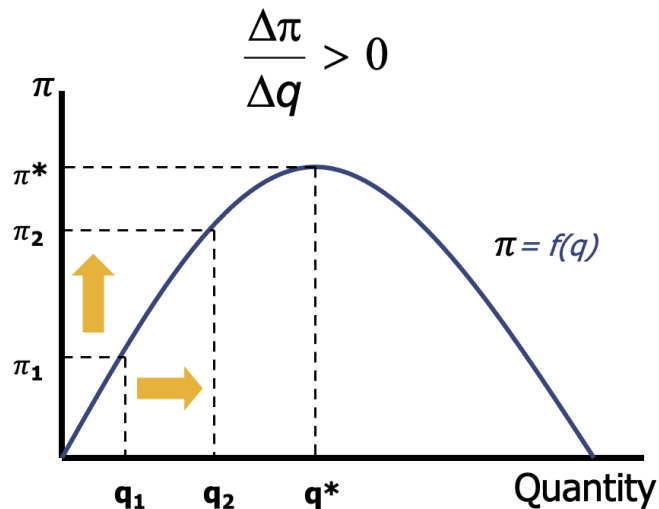
- Mathematization of economics
 - Took off in 1950s (Paul Samuelson, John Hicks, Kenneth Arrow, ...)
 - Some economists are actually mathematicians by training.
 - Anti-Mathematization or highly skeptical of mathematization (e.g., Ronald Coase)
 - Mechanical
 - Lack of economics
 - Tend not to care about the real world
 - The mathematization trend cannot be stopped or reversed
 - Ph.D. program admission requirements include linear algebra, multivariable calculus, real analysis, etc.

Intuition versus Mathematics

- ❑ Excellent math grades are neither necessary nor sufficient to be a great economist.
- ❑ Intuition
 - The act of knowing or sensing without the use of rational processes; immediate cognition.
 - Know something directly without analytic reasoning, bridging the gap between the conscious and nonconscious parts of our mind, and also between instinct and reason.
- ❑ Intuition is extremely valuable.
- ❑ Unfortunately, intuition is difficult (or impossible) to teach. It may be innate.
- ❑ The value of mathematics in economics
 - Intuition may not work all the time
 - Intuition can be wrong (e.g., Diminishing Marginal Utility $\nRightarrow$ Diminishing Marginal Rate of Substitution)
 - Mathematics is precise

Maximization of a function of one variable

- Suppose that a manager of a firm desires to maximize the profits received from selling a particular good.
- Suppose that the profits (π) received depend only on the quantity (q) of the good sold. Mathematically,
 - $\pi = f(q)$.
 - an increase from q_1 to q_2 leads to a rise in π , so keep moving to the right
 - an increase from q^* to q_3 leads to a drop in π , so q^* must be optimum



Derivatives

- Math check
- Question: Consider function $\pi(q) = 2q + \sqrt{q}$
 - Find $\frac{\partial \pi(q)}{\partial q}$
- Value of derivatives at a point

$$\frac{d\pi}{dq} = \frac{df}{dq} = \lim_{h \rightarrow 0} \frac{f(q_1 + h) - f(q_1)}{h}$$

$$\left. \frac{d\pi}{dq} \right|_{q=q_1} > 0$$

$$\left. \frac{d\pi}{dq} \right|_{q=q_3} < 0$$

$$\left. \frac{d\pi}{dq} \right|_{q=q^*} = 0$$

Rules for Finding Derivatives

1. If b is a constant, then $\frac{db}{dx} = 0$

2. If b is a constant, then $\frac{d[bf(x)]}{dx} = bf'(x)$

3. If b is constant, then $\frac{dx^b}{dx} = bx^{b-1}$

4. $\frac{d \ln x}{dx} = \frac{1}{x}$

5. $\frac{da^x}{dx} = a^x \ln a$ for any constant a

A particular case of this rule is $\frac{de^x}{dx} = e^x$

Rules for Finding Derivatives

- Now suppose that $f(x)$ and $g(x)$ are two functions of x and that $f'(x)$ and $g'(x)$ exist. Then:

$$6. \frac{d[f(x) + g(x)]}{dx} = f'(x) + g'(x)$$

$$7. \frac{d[f(x) \cdot g(x)]}{dx} = f(x)g'(x) + f'(x)g(x)$$

$$8. \frac{d\left(\frac{f(x)}{g(x)}\right)}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

provided that $g(x) \neq 0$

Rules for Finding Derivatives

- Finally, if $y = f(x)$ and $x = g(z)$ and if both $f'(x)$ and $g'(z)$ exist, then

$$9. \frac{dy}{dz} = \frac{dy}{dx} \cdot \frac{dx}{dz} = \frac{df}{dx} \cdot \frac{dg}{dz}$$

- This is the *chain rule*. Some examples are

$$10. \frac{de^{ax}}{dx} = \frac{de^{ax}}{d(ax)} \cdot \frac{d(ax)}{dx} = e^{ax} \cdot a = ae^{ax}$$

$$11. \frac{d[\ln(ax)]}{dx} = \frac{d[\ln(ax)]}{d(ax)} \cdot \frac{d(ax)}{dx} = \frac{1}{ax} \cdot a = \frac{1}{x}$$

$$12. \frac{d[\ln(x^2)]}{dx} = \frac{d[\ln(x^2)]}{d(x^2)} \cdot \frac{d(x^2)}{dx} = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$$

First-order condition for a maximum

- If a manager could estimate the function $f(q)$ from some sort of real-world data, it would theoretically be possible to find the point where $df/dq = 0$. At this optimal point (say, q^*),

$$\left. \frac{df}{dq} \right|_{q=q^*} = 0$$

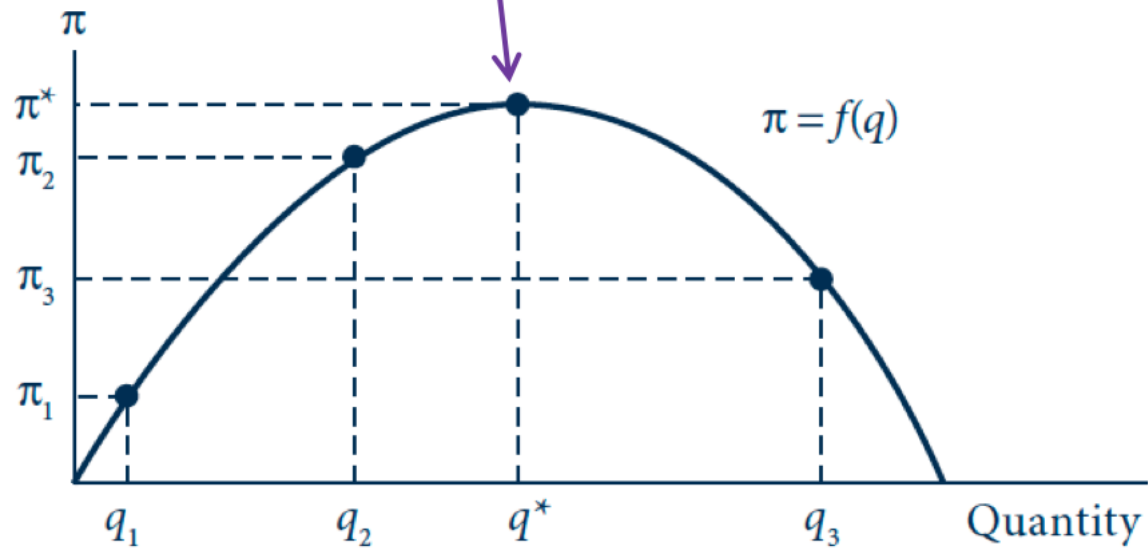
Second-order conditions

- Second derivatives:

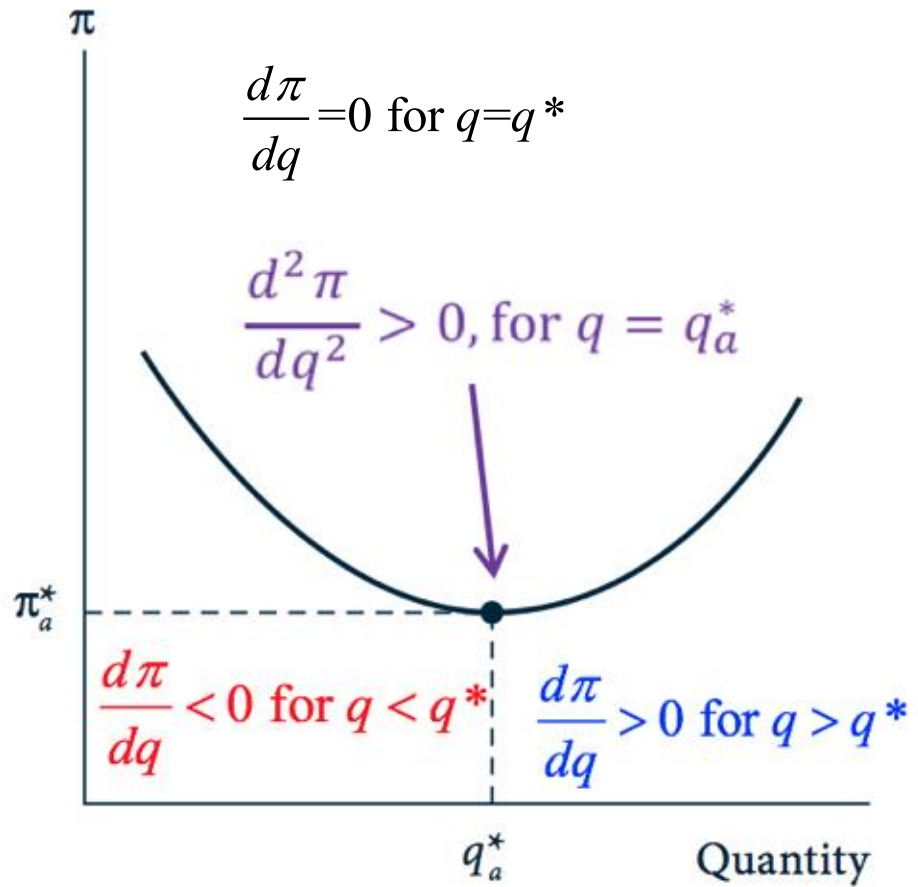
$$\frac{d^2\pi}{dq^2} \text{ or } \frac{d^2f}{dq^2} \text{ or } f''(q).$$

Second-order conditions

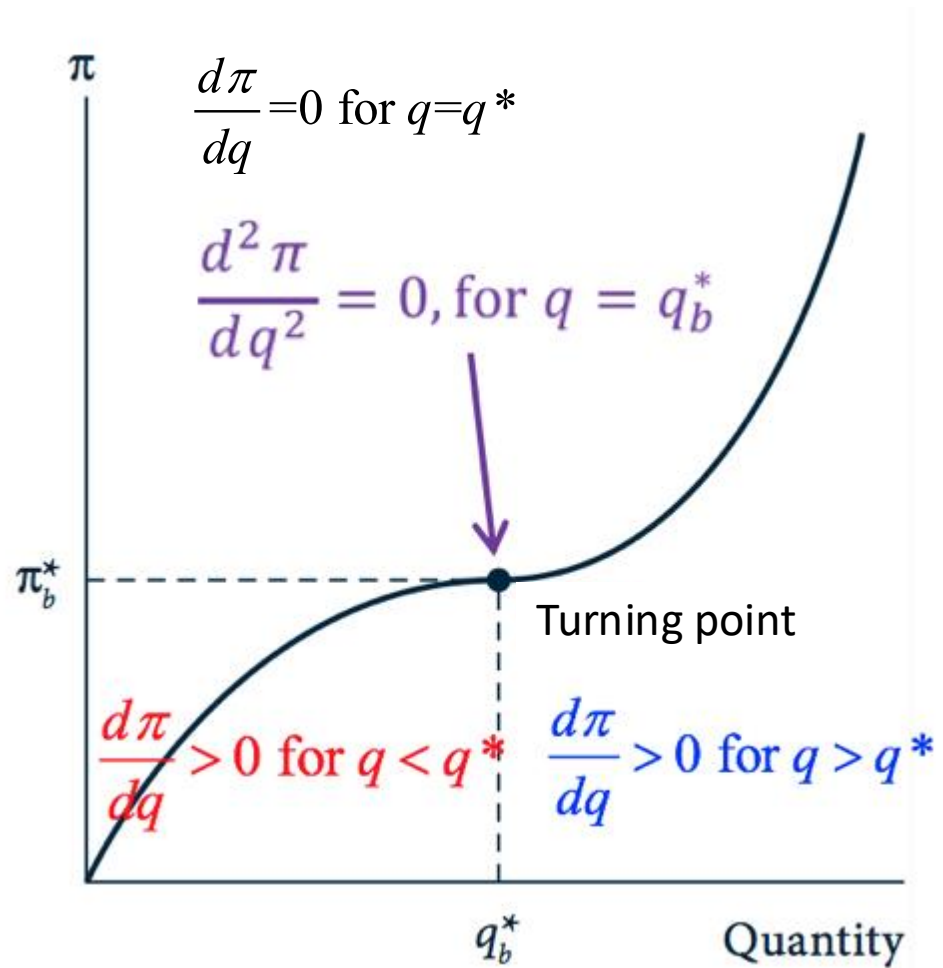
$$\frac{d\pi}{dq} = 0 \text{ for } q = q^*$$
$$\frac{d^2\pi}{dq^2} < 0, \text{ for } q = q^*$$



Second-order conditions



Second-order conditions



Functions of Several Variables

- Economic problems seldom involve functions of only a single variables.
- Most goals of interest to economic agents depend on several variables, and trade-offs must be made among these variables.
- Consumer *utilities* depend on the amount of each good consumed.
- A firm's *production function* depends on the quantity of labor, capital, and land devoted to production.
- This dependence of one variable (y) *on a series of other variables* ($x_1, x_2, \dots, x_n$) is denoted by

$$y = f(x_1, x_2, \dots, x_n).$$

Partial derivatives

- Usually, the only directional slopes of interest are those that are obtained by increasing one of the x 's while holding all the other variables constant.

- The partial derivatives can be denoted by

$$\frac{\partial y}{\partial x_1} \quad \text{or} \quad \frac{\partial f}{\partial x_1} \quad \text{or} \quad f_{x_1} \quad \text{or} \quad f_1$$

- A somewhat more formal definition of the partial derivative is

$$\left. \frac{\partial f}{\partial x_1} \right|_{\bar{x}_2, \dots, \bar{x}_n} = \lim_{h \rightarrow 0} \frac{f(x_1 + h, \bar{x}_2, \dots, \bar{x}_n) - f(x_1, \bar{x}_2, \dots, \bar{x}_n)}{h}$$

Calculating partial derivatives

1. If $y = f(x_1, x_2) = ax_1^2 + bx_1x_2 + cx_2^2$, then

$$\frac{\partial f}{\partial x_1} = f_1 = 2ax_1 + bx_2$$

$$\frac{\partial f}{\partial x_2} = f_2 = bx_1 + 2cx_2.$$

Calculating partial derivatives

2. If $y = f(x_1, x_2) = e^{ax_1 + bx_2}$, then

$$\frac{\partial f}{\partial x_1} = f_1 =$$

$$\frac{\partial f}{\partial x_2} = f_2 =$$

Calculating partial derivatives

3. If $y = f(x_1, x_2) = a \ln x_1 + b \ln x_2$, then

$$\frac{\partial f}{\partial x_1} = f_1 =$$

$$\frac{\partial f}{\partial x_2} = f_2 =$$

Second-order partial derivatives

- Written as $\frac{\partial(\frac{\partial f}{\partial x_j})}{\partial x_i}$, or simply as $\frac{\partial^2 f}{\partial x_j \partial x_i} = f_{ij}$.
- For example, $y = f(x_1, x_2) = a \ln x_1 + b \ln x_2$
 - $f_{11} =$
 - $f_{12} =$
 - $f_{21} =$
 - $f_{22} =$
- Young's theorem: under general conditions, the order in which partial differentiation is conducted to evaluate second-order partial derivatives does not matter. That is

$$f_{ij} = f_{ji}$$

First-order conditions for a maximum

- A necessary condition for a point to be a local maximum is that at this point:

$$f_1 = f_2 = \dots = f_n = 0$$

- A point at which the equation above holds is called a *critically point* of the function.
- It is not necessarily a maximum point unless certain second-order conditions (to be discussed later) hold.

Finding a maximum

- y represents an individual's health (measured on a scale of 0 to 10)
- x_1 and x_2 are daily dosages of two health-enhancing drugs
- Find values of x_1 and x_2 that make y as large as possible.
- Suppose that y is function of x_1 and x_2 given by
$$y = -(x_1 - 1)^2 - (x_2 - 2)^2 + 10$$

Taking partial derivatives of y with respect to x_1 and x_2 and applying necessary conditions of finding a maximum yields

$$\begin{aligned}\frac{\partial y}{\partial x_1} &= -2x_1 + 2 = 0 \\ \frac{\partial y}{\partial x_2} &= -2x_2 + 4 = 0\end{aligned}$$

Or

$$\begin{aligned}x_1^* &= 1, \\ x_2^* &= 2.\end{aligned}$$

Elasticity – A general definition

- Economists use elasticities A LOT.
- Elasticities focus on the proportional effect of a change in one variable on another.
- Unit-free – the units “cancel out”
- Suppose that y is a function of x : $y(x)$
- Then the elasticity of y with respect to x , denoted by $e_{y,x}$, is defined

$$\text{as } e_{y,x} = \frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \frac{\Delta y}{\Delta x} \cdot \frac{x}{y} = \frac{\partial y}{\partial x} \cdot \frac{x}{y}$$

Elasticity

1. If $y = a + bx + \text{other terms}$,

In this case,

$$e_{y,x} = \frac{\partial y}{\partial x} \cdot \frac{x}{y} = b \cdot \frac{x}{y} = b \cdot \frac{x}{a + bx + \dots}$$

Elasticity

2. If $y = ax^b$ (exponential form),

$$e_{y,x} = \frac{dy}{dx} \cdot \frac{x}{y} = abx^{b-1} \cdot \frac{x}{ax^b} = b$$

The elasticity is a constant, independent of where it is measured.

Elasticity

A logarithmic transformation of this equation ($y = ax^b$) provides a convenient alternative definition of elasticity.

$$\ln y = \ln a + b \ln x$$

$$e_{y,x} = \frac{d \ln y}{d \ln x} = b$$

“logarithmic transformation” is frequently used as it is the easiest way of proceed in calculating elasticities.

The chain rule

- If $y = f[x_1(a), x_2(a), x_3(a)]$

$$\frac{dy}{da} = \frac{\partial f}{\partial x_1} \cdot \frac{dx_1}{da} + \frac{\partial f}{\partial x_2} \cdot \frac{dx_2}{da} + \frac{\partial f}{\partial x_3} \cdot \frac{dx_3}{da}$$

- Example:
 - Pizza 1 costs p per pie; pizza 2 costs $2p$ per pie; pizza 3 costs $3p$ per pie.
 - The consumer wants to spend \$30 to each type of pizza.
 - Question: how will the underlying price p affect the total number of pizza purchased?

$$x_1 = 30/p, \quad x_2 = 30/2p, \quad x_3 = 30/3p.$$

$$\text{Total pizza purchased (y)} = f[x_1(p), x_2(p), x_3(p)] = x_1(p) + x_2(p) + x_3(p)$$

Applying the chain rule, we get

$$\frac{dy}{dp} = f_1 \cdot \frac{dx_1}{dp} + f_2 \cdot \frac{dx_2}{dp} + f_3 \cdot \frac{dx_3}{dp} = -30p^{-2} - 15p^{-2} - 10p^{-2} = -55p^{-2}$$

Question:

Suppose that initially $p = 5$, what happens to total pizza purchases if p increases by 0.05?

Implicit function theorem

- Suppose that $z = f(x, y) = c$, which is a constant.
- If we hold the value of z constant, we have created an implicit relationship between x and y ($y = g(x)$) showing how changes in them must be related to keep the value of the function constant.
 - $z = f(x, g(x)) = c$
- Using the chain rule to differentiate the relationship with respect to x yields:
 - $0 = f_x + f_y \cdot \frac{dg(x)}{dx}$
 - $\frac{dg(x)}{dx} = \boxed{\frac{dy}{dx} = -\frac{f_x}{f_y}}$

Implicit function theorem

- A production possibility frontier for two goods of the form:

$$2x^2 + y^2 = 225$$

- Can be rewritten as:

$$f(x, y) = 2x^2 + y^2 - 225 = 0$$

- the trade-off between x and y is

$$\frac{dy}{dx} = \frac{-f_x}{f_y} = \frac{-4x}{2y} = \frac{-2x}{y}$$

The Envelope Theorem

- It is a major application of the idea of implicit functions.
- It is about how the optimal value for a particular function changes when a parameter of the function changes.
 - The effects that changing the market price of a commodity will have on an individual's purchases
- Illustrate through an example...

A specific example of the Envelope Theorem

- Suppose that y is a function of x

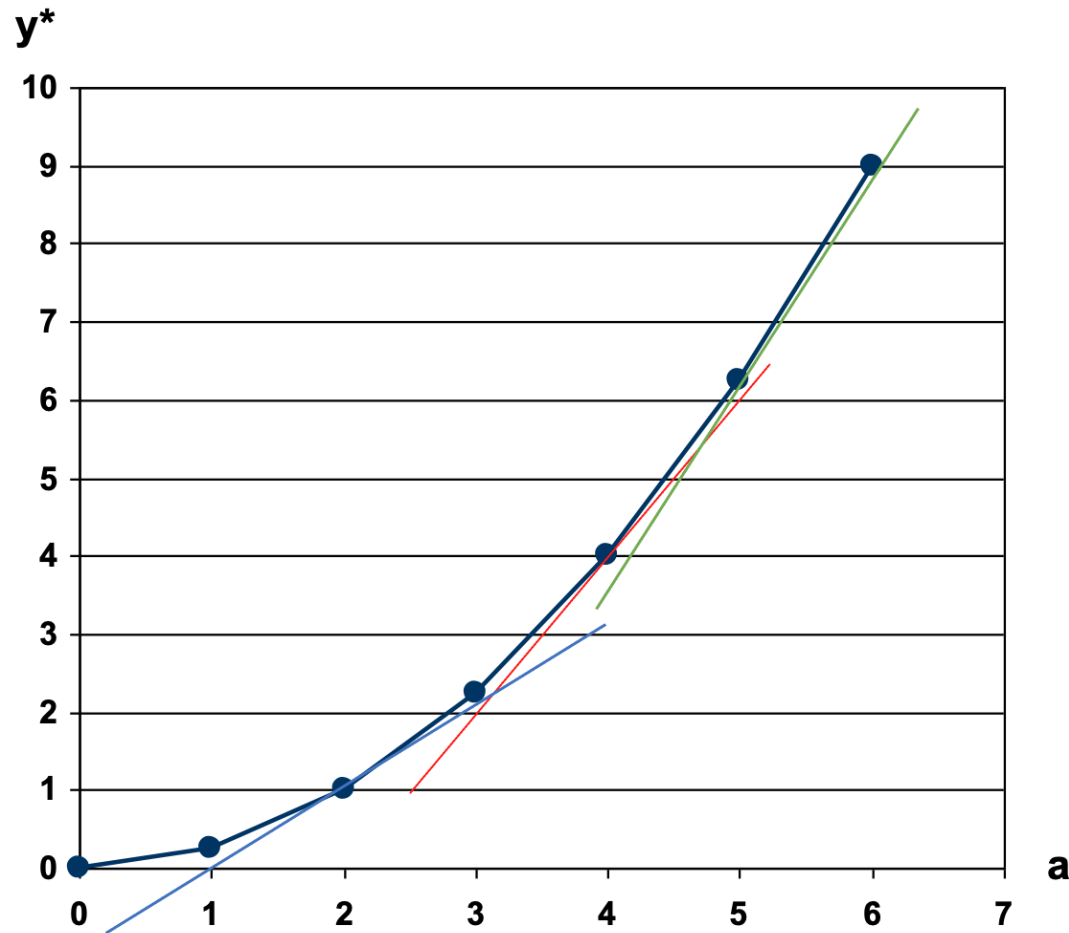
$$y = -x^2 + ax$$

- For different values of a , this function represents a family of inverted parabolas.
- If a is assigned a specific value, then y becomes a function of x only and the value of x that maximizes y can be calculated:

Optimal values of y and x for alternative values of a in $y = -x^2 + ax$

<u>Value of a</u>	<u>Value of x^*</u>	<u>Value of y^*</u>
0	0	0
1	1/2	1/4
2	1	1
3	3/2	9/4
4	2	4
5	5/2	25/4
6	3	9

Illustration of the Envelope theorem



- As a increases, the maximal value for y (y^*) increases.
- The relationship between a and y^* is quadratic.
- This figure also shows that the slope of the curve is positive and increases in a .

The direct, but maybe time-consuming approach

- We are interested in how y^* changes when a changes.
- First, we can calculate the slope of the function $y = -x^2 + ax$ directly.
 - Find x^* as a function of a
- Then, substituting this value of x^* in $y = -x^2 + ax$ gives y^* as a function of a
- Lastly, calculate $\frac{dy^*}{da}$.

The envelope shortcut

- For small changes in a , dy^*/da can be computed by holding x *at its optimal value* and simply calculating $\frac{\partial y}{\partial a}$ from the objective function directly.

$$\begin{aligned}\frac{dy^*}{da} &= \left. \frac{\partial y}{\partial a} \right|_{x=x^*(a)} = \left. \frac{\partial y}{\partial x} \cdot \frac{\partial x}{\partial a} \right|_{x=x^*(a)} + \left. \frac{\partial f}{\partial a} \right|_{x=x^*(a)} = 0 + \left. \frac{\partial f}{\partial a} \right|_{x=x^*(a)} \\ &= \left. \frac{\partial(-x^2 + ax)}{\partial a} \right|_{x=x^*(a)} = x^*(a)\end{aligned}$$

- Substitute in $x^* = a/2$,

$$\frac{dy^*}{da} = x^*(a) = a/2$$

The Envelope Theorem

- The Envelope Theorem states that the change in the optimal value of a function with respect to a parameter of that function
-- can be found by **partially differentiating** the objective function while holding x (or several x 's) at its optimal value

$$\frac{dy^*}{da} = \frac{\partial y}{\partial a} \{x = x^*(a)\}$$

- The notation reminds that $\frac{\partial y}{\partial a}$ must be computed at the optimal value of x .

Constrained maximization

- So far we focus on finding the maximum value of y without restricting the choices of the x 's available.
- However,
 - Managers choose output x to maximize profit $y \rightarrow$ need all x 's positive
 - Choices are constrained by purchasing power (budget constraint)
- y may not be as large as it could be

Lagrange multiplier method

- Introduce the Lagrange multiplier as an additional variable to be solved corresponding to the constraint on the x 's.

- The formal problem:

- Find the values of $x_1, x_2, \dots, x_n$ that maximize

$$y = f(x_1, x_2, \dots, x_n)$$

Subject to a constraint on x 's

$$g(x_1, x_2, \dots, x_n) = 0$$

Set up the Lagrangian expression

$$\mathcal{L} = f(x_1, x_2, \dots, x_n) + \lambda g(x_1, x_2, \dots, x_n)$$

- λ is the Lagrange multiplier.
- When the constraint holds (or, the constraint is binding), $\mathcal{L} = f$, because $g(x_1, x_2, \dots, x_n) = 0$.
- If we restrict to only the values of the x 's that satisfy the constraint,
- Finding the constrained maximum value of f is equivalent to finding a critical value of $\mathcal{L}$

First-order conditions

$$\begin{array}{l} d\mathcal{L}/dx_1 = f_1 + \lambda g_1 = 0 \\ d\mathcal{L}/dx_2 = f_2 + \lambda g_2 = 0 \\ \quad \cdot \\ \quad \cdot \\ \quad \cdot \\ d\mathcal{L}/dx_n = f_n + \lambda g_n = 0 \\ d\mathcal{L}/d\lambda = g(x_1, x_2, \dots, x_n) = 0 \end{array} \left. \vphantom{\begin{array}{l} d\mathcal{L}/dx_1 = f_1 + \lambda g_1 = 0 \\ d\mathcal{L}/dx_2 = f_2 + \lambda g_2 = 0 \\ \quad \cdot \\ \quad \cdot \\ \quad \cdot \\ d\mathcal{L}/dx_n = f_n + \lambda g_n = 0 \\ d\mathcal{L}/d\lambda = g(x_1, x_2, \dots, x_n) = 0 \end{array}} \right\} \begin{array}{l} n+1 \text{ equations for} \\ n+1 \text{ unknown} \\ \text{variables} \end{array}$$

The solution will have **two properties**:

1. the x 's will **obey** the constraint
2. these x 's will make the value of L (and **therefore f**) **as large as possible**

Interpretation of the Lagrange multiplier

- The Lagrange multiplier is more than just a math “trick”.
- Important economic interpretation:
- Rewrite the first-order conditions as

$$\frac{f_1}{-g_1} = \frac{f_2}{-g_2} = \dots = \frac{f_n}{-g_n} = \lambda$$

- the numerators = the *marginal benefit* that one more unit of x_i will have for the function f
- the denominators = the added burden (or *marginal cost*) on the constraint of using more x_i

Interpretation of the Lagrange multiplier

- At the optimal choices for the x 's, the ratio of the marginal benefit of increasing x_i to the marginal cost of increasing x_i **should be the same for every x**
- λ is the **common cost-benefit ratio** for all of the x 's

$$\lambda = \frac{\text{marginal benefit of } x_i}{\text{marginal cost of } x_i}$$

Lagrange multiplier = shadow price

- The Lagrangian multiplier, provides a measure of how the ***relaxation in the constraint*** will affect the value of y
- λ is called a “***shadow price***” to the constraint:
 - A high value of $\lambda \rightarrow y$ could be increased substantially by relaxing the constraint
 - A low value of $\lambda \rightarrow$ there is not much to be gained by relaxing the constraint
 - $\lambda=0$ implies that the constraint is *not binding* $\rightarrow$ same as solving the unconstrained maximization problem

An example: optimal fences and constrained maximization

- Suppose a farmer had a certain length of fence (P) and wished to enclose the largest possible rectangular area
 - Let x be the length of one side
 - Let y be the length of the other side
- What shape area should the farmer choose?
- The problem is to choose x and y to maximize $A = x \cdot y$, subject to the constraint that the perimeter is fixed at $P = 2x + 2y$.

Duality

- Any **constrained maximization** problem has associated with it a **dual** problem in **constrained minimization** that focuses attention on the constraints in the original problem

Original Problem	Dual Problem
• Firms minimize the cost of inputs to produce a given level of output	• Firms maximize output for a given cost of inputs purchased
• Individuals maximize utility subject to a budget constraint	• Individuals minimize the expenditure needed to achieve a given level of utility

Still that example

- What is the dual problem in words?
- How to set up the math problem?
 - Minimizes what?
 - Subject to what?
- How to set up the Lagrange expression?
 - First-order conditions?
 - Solve for x and y and λ
 - How would you interpret this λ ?

Second-order conditions and curvature

- We now discuss the sufficient conditions for an optimum and their relationship to second-order conditions.
- We will also discuss the economic explanations for these curvature conditions.
- Functions of two variables x_1 and x_2 .

$$y = f(x_1, x_2)$$

→ Ensure that the dy is decreasing for all movements through the critical point

Unconstrained maximization

- A formal analysis:

Total differential of the function is

$$dy = f_1 dx_1 + f_2 dx_2$$

The total differential of dy function is

$$d^2y = (f_{11}dx_1 + f_{12}dx_2)dx_1 + (f_{21}dx_1 + f_{22}dx_2)dx_2$$

$$\text{Or } d^2y = f_{11}dx_1^2 + f_{12}dx_2dx_1 + f_{21}dx_1dx_2 + f_{22}dx_2^2$$

By Young's theorem, $f_{12} = f_{21}$,

Rearrange terms,

$$d^2y = f_{11}dx_1^2 + 2f_{12}dx_1dx_2 + f_{22}dx_2^2$$

$$d^2y = f_{11}dx_1^2 + 2f_{12}dx_1dx_2 + f_{22}dx_2^2$$

- For this equation to be unambiguously negative for any change in the x 's, f_{11} and f_{22} *must be negative*.

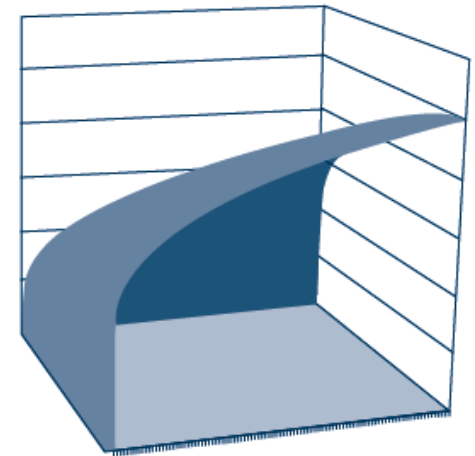
- If $dx_2 = 0$, then $d^2y = f_{11}dx_1^2$
 - for $d^2y < 0$, it must be $f_{11} < 0$
- If $dx_1 = 0$, then $d^2y = f_{22}dx_2^2$
 - for $d^2y < 0$, it must be $f_{22} < 0$
- If neither dx_1 nor dx_2 is 0, then d^2y is unambiguously negative if and only if

$$f_{11}f_{22} - f_{12}^2 > 0$$

the second partial derivatives (f_{11} and f_{22}) must be sufficiently negative so that they outweigh any possible perverse effects from the cross-partial derivatives ($f_{12} = f_{21}$)

Concave functions

- Functions that obey such a condition are called (strictly) *concave functions*.
 - Functions with one variable: $f''(x) < 0$
 - Functions with two variables: $f_{11} < 0$ (and $f_{22} < 0$) , $f_{11}f_{22} - f_{12}^2 > 0$
 - Functions with multiple variables: Hessian matrix is negative definite (not required in this course)
- In three dimensions, concave functions resemble inverted teacups.



(a) $k = 0.2$

Constrained maximization

- Choose x_1 and x_2 to maximize

$$y = f(x_1, x_2)$$

- Subject to the linear constraint

$$c - b_1x_1 - b_2x_2 = 0$$

- We can set up the Lagrangian

$$L = f(x_1, x_2) + \lambda(c - b_1x_1 - b_2x_2)$$

Constrained maximization

- First-order conditions:

$$\begin{aligned}f_1 - \lambda b_1 &= 0 \\f_2 - \lambda b_2 &= 0 \\c - b_1x_1 - b_2x_2 &= 0\end{aligned}$$

- To ensure that the point is a local maximum, we need to use the “second” total differential:

$$d^2y = f_{11}dx_1^2 + 2f_{12}dx_1dx_2 + f_{22}dx_2^2 < 0$$

Constrained maximization

- Because of the constraint, not all possible small changes in the x 's are permissible.
- Only those that satisfy the constraint can be considered valid alternatives to the critical point.
- We must calculate the total differential of the constraint:
$$-b_1dx_1 - b_2dx_2 = 0$$
$$dx_2 = -(b_1/b_2)dx_1$$
- These are allowable relative changes in x_1 and x_2 .

Constrained maximization

We know $f_1/f_2 = b_1/b_2$ from FOC, combining with the equation above yields

$$dx_2 = -(f_1/f_2) dx_1$$

We can take dx_2 into d^2y expression,

$$d^2y = f_{11}dx_1^2 - 2f_{12}(f_1/f_2)dx_1^2 + f_{22}(f_1^2/f_2^2)dx_1^2$$

Combining terms gives

$$d^2y = (f_{11}f_2^2 - 2f_{12}f_1f_2 + f_{22}f_1^2)[dx_1^2/f_2^2]$$

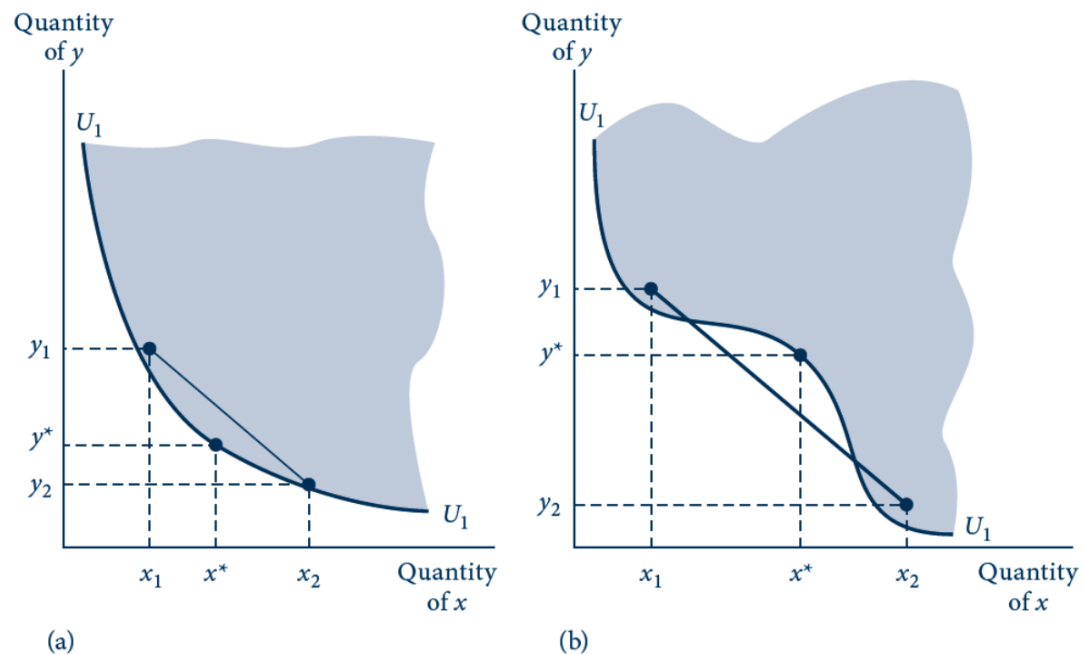
Consequently, for $d^2y < 0$, it must be the case that

$$f_{11}f_2^2 - 2f_{12}f_1f_2 + f_{22}f_1^2 < 0$$

Functions satisfy this condition are called *quasi-concave functions*.

Quasi-concave functions

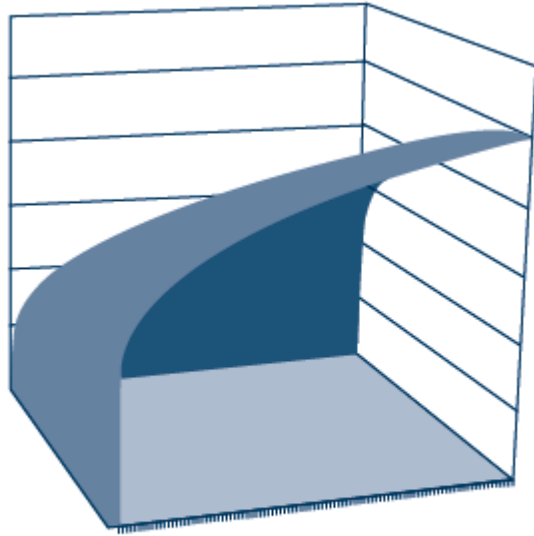
- The set of all points for which such a function takes on a value greater than any specific constant is a convex set.
- i.e., any two points in the set can be joined by a line contained completely within the set.



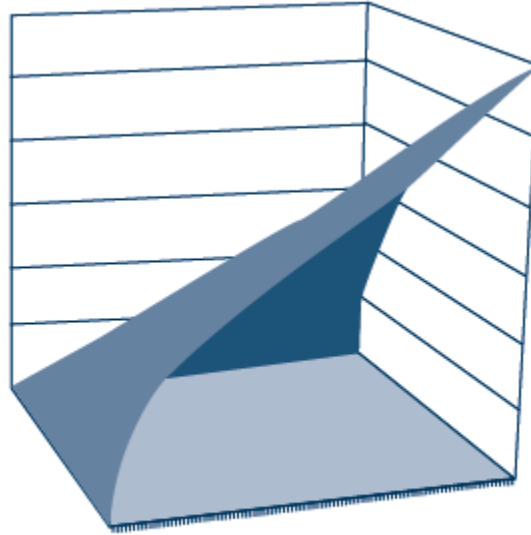
Concave and Quasi-Concave Functions

- $y = f(x_1, x_2) = (x_1 \cdot x_2)^k$
 - Where $x_1 > 0, x_2 > 0$, and $k > 0$
 - No matter what value k takes, this function is quasi-concave
 - $f_{11}f_2^2 - 2f_{12}f_1f_2 + f_{22}f_1^2 < 0$
- Whether or not the function is concave depends on the value of k
 - If $k < 0.5$, the function is concave
 - If $k > 0.5$, the function is convex
 - For points where $x_1 = x_2$,
$$y = (x_1^2)^k = x_1^{2k}$$

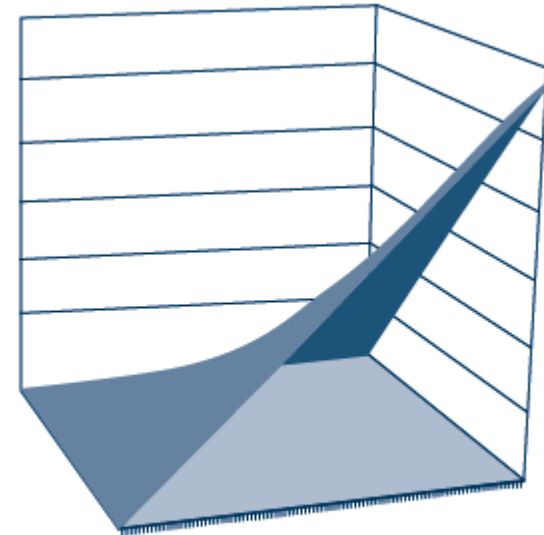
Concave and Quasi-Concave Functions



(a) $k = 0.2$



(b) $k = 0.5$



(c) $k = 1.0$

In all three cases these functions are quasi-concave.

But only for $k = 0.2$ is the function strictly concave.

The case $k = 1.0$ clearly shows nonconcavity because the function is not below its tangent plane.